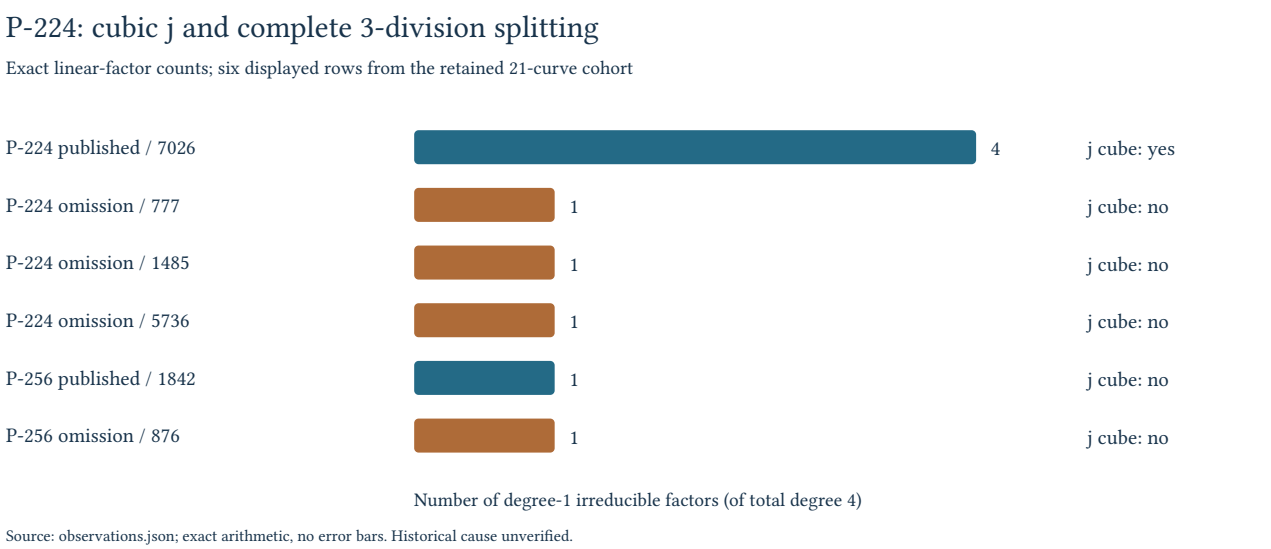
P-224 cubic j and 3-division splitting

Observation record / 25 October 2026

Historical selection criterion: unresolved. This report preserves a specific lead from a fixed comparison of 21 curves: ten publications and eleven reconstructed omissions across six streams.

For $y^2 = x^3 - 3x + b$ over F_p, $j = \frac{6912}{4-b^2}$ modulo p. The published P-224 j-value is a cube in its base field. The three omitted P-224 j-values are not. The published P-224 3-division polynomial splits into four linear factors; each omission factors into degrees 1 and 3.



Required counterexample: published P-256 has a non-cubic j-value, as does its omitted block 876. Both have the factor pattern 1+3. A universal cube-root requirement therefore fails this comparison.

The 3-division polynomial, after removing its constant scalar, is $x^4 - 6x^2 + 4bx - 3$. Its roots describe nonzero 3-torsion x-coordinates. This observation does not establish a transfer attack on the prime subgroup.

Replay and exact identities

The adjacent inputs.json retains all 21 full curve identities, field characteristics, coefficients, subgroup orders, generators, cofactors, seeds, timestamps and ordinals. observations.json retains the exact j-values and polynomial factor degrees. replay.py independently computes j using Python and native PARI/GP, checks all factor products and p/n primality, regenerates all seeds, and checks coefficient/identity bindings.

Command: `python3 replay.py --gp gp`

The replay passed all 21 rows and two tampering controls. It does not rerun SEA, authenticate the original historical program or promote a research claim.

Interpretation boundary and next check

Observed: the P-224 publication differs from each of its three omissions in cubic-residue j and 3-division splitting.

Exploratory branch hypothesis: when $p = 1$ modulo 4, require a cube root of j. P-224 is the only such field here. This condition preserves all publications and rejects these three omissions, leaving eight unexplained. It is not an independent cross-field prediction.

Unverified historical mechanism: a field-dependent point-counting or validation path could require that root and treat failure as rejection. No original caller, configuration, fallback or failure handling has been recovered. The observed pattern alone does not prove such a path existed.

Falsifiable next investigation

Find a dated implementation and its actual caller. Identify branch admission, cube-root failure handling, fallback and return status. Replay the declared path on the complete cohort, including non-cubic published P-256. A normal fallback would weaken that implementation hypothesis. A matching P-224 rejection would retain a branch-specific lead but would still leave the other omissions to explain.

Custody and limitations

source-provenance.json binds the original exact-input and j/splitting records by SHA-256. These are reconstructed omissions, not documented rejection logs. No official experiment or goal status changes. No timing, attack-cost or weakness claim is made. No canonical isogeny graph changes because this study proves no new map or subgroup-transport edge.

Sources

Internal exact-input and arithmetic receipts, retrieved and replayed 25 October 2026, are identified in source-provenance.json.

Original seed reconstruction: the omission cause is explicitly left unresolved.

Sage finite-field curve documentation: j and twists do not in general determine base-field isomorphism.

Full details, controls and paths: report.md. This PDF is a derived view of the adjacent machine-readable record, not historical provenance.